Empirical Forecast Evaluation & Baseline Verification

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1. Executive Abstract & Scope

This paper establishes the out-of-sample empirical evaluation of the PredictaStock 4-Pillar Volatility Scenario Model over forward 5-trading-day horizons across front-line Indian liquid equities (Nifty 50 constituents). To guarantee scientific integrity, the model is strictly benchmarked against three non-trivial baseline models: (1) the No-Change Model, (2) the Nifty 50 Passive Buy & Hold Benchmark, and (3) a 20-DMA Trend Extrapolation Baseline. All forward evaluations deduct an institutional statutory friction of 0.15% roundtrip.

2. Information Filtration & Strict Leakage Control

In quantitative finance, look-ahead bias invalidates historical claims. PredictaStock enforces an information filtration boundary F_t where features, moving averages, and indicators at trade inception session t utilize only completed daily bars on or before session t. Forward returns are evaluated at the exact closing price of the 5th trading session (t+5). Original predictions are stored with unique run_id hashes, guaranteeing idempotency.

3. Indian Regulatory Transaction Friction Model (0.15% Roundtrip)

Cost Component	Statutory Rate	Regulatory Basis
Securities Transaction Tax (STT)	0.100%	Delivery turnover (Buy & Sell)
Brokerage (Discount Institutional)	0.050%	Turnover capped at Rs. 20/order
Exchange Turnover Charges	0.00345%	NSE exchange fee
Stamp Duty (State Stamp Act)	0.015%	Buy delivery turnover
Goods & Services Tax (GST)	18.00%	On Brokerage + Exchange fees
Execution Slippage Allowance	0.050%	Bid-ask spread absorption
Total Net Deducted Friction	**0.150%**	**Deducted from every 5-day return**

4. Out-of-Sample Empirical Results (4-Model Matrix)

Strategy / Benchmark Model	Sample N	Hit Rate	Avg Return	Alpha vs Nifty	Profit Factor	Max DD
PredictaStock Multi-Factor (Model)	**85**	**42.4%**	**-0.22%**	**-0.07%**	**0.81**	**-36.56%**
Nifty 50 Buy & Hold Benchmark	85	35.3%	-0.15%	0.00% (Base)	N/A	N/A
20-DMA Trend Extrapolation Baseline	85	45.9%	-0.23%	-0.08%	N/A	N/A
No-Change Model (0% Return)	85	30.6%	0.00%	+0.15%	N/A	N/A

5. Market Regime Sensitivity Analysis

Disaggregating performance across macro market regimes highlights structural asymmetries:

* Bull Market Sessions (Nifty 50 Return > 0): Directional Hit Rate = 53.3%.
* Bear Market Sessions (Nifty 50 Return <= 0): Directional Hit Rate = 36.4%.

In constructive sessions, momentum and fundamental screens align with market continuation. In bearish sessions, downside volatility spikes degrade individual stock alpha, proving that disciplined risk management and stop-loss adherence are paramount.

6. Statutory Disclaimers & Governance Notice

1. Educational & Decision-Support Only: PredictaStock is designed as a structured research dashboard. It does not execute automated broker orders or guarantee returns.

2. Non-SEBI Registration: Not registered with SEBI under SEBI (Research Analysts) Regulations, 2014. Forecasted volatility cones represent bounded statistical dispersion scenarios, not guaranteed trading advice.